

# THQ #3

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## Question 1 (Sec 3.1)

Put the following problems into standard form.

(a)

Objective Function:

$$\mathbf{\max} \ 3x_1 - 2x_2$$

Constraints:

- $5x_1 + 2x_2 - 3x_3 + x_4 \leq 7$
- $3x_2 - 4x_3 \leq 6$
- $x_1 + x_3 - x_4 \geq 11$
- $x_1, x_2, x_3, x_4 \geq 0$       **(non-negative)**

Ok so I understand that max has to be turned into a minimum, and we need to create slack variables to help turn these inequalities to equalities.

Solved:

Objective Function:

$$\mathbf{\min} \ -3x_1 + 2x_2$$

Constraints:

- $5x_1 + 2x_2 - 3x_3 + x_4 + s_1 = 7$
- $3x_2 - 4x_3 + s_2 = 6$
- $x_1 + x_3 - x_4 - s_3 = 11$
- $x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0$       **(non-negative)**

(b)

**Objective Function:**

$$\mathbf{min} \ x_2 + x_3 + x_4$$

**Constraints:**

- $x_1 + x_2 \geq 6$
- $x_2 + x_3 - x_4 \leq 1$
- $5x_1 - 6x_2 + 7x_3 - 8x_4 \geq 2$
- $x_1 \geq 0, x_2 \leq 0, x_3, x_4$  unrestricted

Since the obj function is minimized, I'll leave it alone. For inequalities, I'll add some slack variables as needed, and for the unrestricted variables, I'll separate each of them to its positive and negative, and make sure to show that everywhere. Since  $x_2$  is negative, I need to make it positive as well, multiplying all  $x_2$  terms by -1.

**Solved:**

**Objective Function:**

$$\mathbf{min} \ -x_2 + x'_3 - x''_3 + x'_4 - x''_4$$

**Constraints:**

- $x_1 - x_2 - s_1 = 6$
- $-x_2 + x'_3 - x''_3 - x'_4 + x''_4 + s_2 = 1$
- $5x_1 + 6x_2 + 7x'_3 - 7x''_3 - 8x'_4 + 8x''_4 - s_3 = 2$
- $x_1, x_2, x'_3, x''_3, x'_4, x''_4, s_1, s_2, s_3 \geq 0$  **(non-negative)**

## Question 2 (Sec 3.2)

**Objective Function:**

$$\mathbf{min} \ 10x_1 + 4x_2$$

**Constraints:**

- $3x_1 + 2x_2 - x_3 = 60$
- $7x_1 + 2x_2 - x_4 = 84$
- $3x_1 + 6x_2 - x_5 = 72$

- $x_1, x_2, x_3, x_4, x_5 \geq 0$  (non-negative)

Simplex Algorithm:

Simplex Algorithm

Step 0: Transform

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | Rhs |
|-------|-------|-------|-------|-------|-------|-----|
| $x_2$ | 3     | 2     | -1    | 0     | 0     | 60  |
| $x_1$ | 7     | 2     | 0     | -1    | 0     | 84  |
| $x_5$ | 3     | 6     | 0     | 0     | -1    | 72  |
|       | 10    | 4     | 0     | 0     | 0     | 0   |

$\Downarrow R_4 = -R_3$

|       |    |    |    |    |   |     |
|-------|----|----|----|----|---|-----|
| $x_2$ | 3  | 2  | -1 | 0  | 0 | 60  |
| $x_1$ | 7  | 2  | 0  | -1 | 0 | 84  |
| $x_5$ | -3 | -6 | 0  | 0  | 1 | -72 |
|       | 10 | 4  | 0  | 0  | 0 | 0   |

$\Downarrow$   
 $R_4 = R_1$   
 $R_2 = 2R_4$   
 $R_3 = 6R_4$   
 $R_5 = 4R_4$

|       |               |   |                |    |    |      |
|-------|---------------|---|----------------|----|----|------|
| $x_2$ | $\frac{3}{2}$ | 1 | $-\frac{1}{2}$ | 0  | 0  | 30   |
| $x_1$ | 4             | 0 | -1             | -1 | 0  | 24   |
| $x_5$ | 6             | 0 | -3             | 0  | -1 | 108  |
|       | 4             | 0 | 2              | 0  | 0  | -120 |

$\Downarrow$   
 $R_4 = R_2$   
 $R_1 = \frac{3}{2}R_4$   
 $R_3 = 6R_4$   
 $R_5 = 4R_4$

|       |   |   |                |                |   |      |
|-------|---|---|----------------|----------------|---|------|
| $x_2$ | 0 | 1 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 0 | 21   |
| $x_1$ | 1 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 0 | 6    |
| $x_5$ | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 1 | 72   |
|       | 0 | 0 | 1              | 1              | 0 | -120 |

**Solution:**

$$x_3 + x_4 = -144 + z \implies z = 144 \quad \text{at } (6, 21, 0, 0, 72)$$

### Question 3 (Sec 3.3)

Use the simplex method covered in this class (with simplex tableau) to solve the following LP model, using  $x_1$  and  $x_2$  as the initial BV.

**Objective Function:**

$$\min x_1 + x_2 + 4x_3 + 7x_4$$

**Constraints:**

- $x_1 + x_2 + 5x_3 + 2x_4 = 8$
- $2x_1 + x_2 + 8x_3 = 14$
- $x_1, x_2, x_3, x_4 \geq 0$  (non-negative)

**Simplex Algorithm:**

The image shows handwritten work for the simplex method. It includes three tableaux and the necessary row operations.

**Initial Tableau:**

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | RHS |
|-------|-------|-------|-------|-------|-----|
| $x_1$ | 1     | 1     | 5     | 2     | 8   |
| $x_2$ | 2     | 1     | 8     | 0     | 14  |
| $Z$   | 1     | 1     | 4     | 7     | 0   |

Row operations:  $R_2 - 2R_1$ ,  $R_2 - R_1$

**Second Tableau:**

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | RHS |
|-------|-------|-------|-------|-------|-----|
| $x_1$ | 1     | 1     | 5     | 2     | 8   |
| $x_2$ | 0     | -1    | -2    | -4    | -2  |
| $Z$   | 0     | 0     | -1    | 5     | -8  |

Row operations:  $R_2 \times -1$ ,  $R_1 - R_2$ ,  $R_2 - 0R_1$

**Third Tableau:**

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | RHS |
|-------|-------|-------|-------|-------|-----|
| $x_1$ | 1     | 0     | 3     | -2    | 6   |
| $x_2$ | 0     | 1     | 2     | 4     | 2   |
| $Z$   | 0     | 0     | -1    | 9     | -8  |

Row operations:  $R_1 - 3R_2$ ,  $R_2 - 0R_1$

**Final Tableau:**

|       | $x_1$ | $x_2$ | $x_3$ | $x_4$ | RHS |
|-------|-------|-------|-------|-------|-----|
| $x_1$ | 1     | 0     | 0     | -8    | 3   |
| $x_2$ | 0     | 1     | 0     | 2     | 1   |
| $Z$   | 0     | 0     | 0     | 7     | -7  |

Row operations:  $R_1 + 8R_2$ ,  $R_2 - 0R_1$

Handwritten notes include: "BV =  $\frac{6}{3} = 2$ ", "Enter  $x_3$  & Exit  $x_2$ ", and "BV =  $\frac{2}{2} = 1$ ".

**Solution:**

$$\frac{1}{2}x_2 + 7x_4 = -7 + z \implies z = 7 \quad \text{at } (3, 0, 1, 0)$$